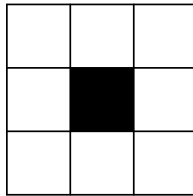


ARML Local 2017 Team Problems

- T-1. A fair six-sided die has faces with values 0, 0, 1, 3, 6, and 10. Compute the smallest positive integer that *cannot* be the sum of four rolls of this die.
- T-2. For a positive integer k , let z_k be the number of terminal zeroes of the product $1!2!\cdots k!$. For example, $z_6 = 2$ because $1!2!3!4!5!6! = 24883200$. Compute z_{100} .
- T-3. Let $ARML$ be a square of side length 5. A point B on side $\overline{MR}$ and C on side $\overline{ML}$ are selected uniformly at random and independent of one another. Compute the expected area of triangle ABC .
- T-4. The edges of regular hexagon $ABCDEF$ are made of mirrors. A laser is fired from A toward the interior of edge $\overline{CD}$, striking it at point G . The laser beam reflects off the interior of exactly one additional edge and returns to A . Compute $\tan(\angle DAG)$.
- T-5. Let S be the set of lattice points $\{(x, y) : x, y \in \mathbb{Z}, 0 \leq x \leq 3, 0 \leq y \leq 4\}$. Compute the number of subsets of S of 4 lattice points that form the vertices of a square.
- T-6. If A is an acute angle such that $\sin 15^\circ + \cos 15^\circ = \sqrt{2} \sin A$, compute $\cos A$.
- T-7. Consider the following algorithm for a given non-negative integer n :
 Step 1: Initialize $k = 1$.
 Step 2: Replace n with the product of its digits.
 Step 3: If $n \leq 9$ return k and exit. If $n > 9$, increment k by 1 and return to Step 2.
 For example, if $n = 125$, then the algorithm returns 2 because 125 is replaced by $1 \cdot 2 \cdot 5 = 10$ which is replaced by $1 \cdot 0 = 0$, at which point the algorithm returns 2 and exits. Compute the smallest value of n such that this algorithm returns 3.
- T-8. Three co-planar squares, $BAHT$, $CAIN$, and $BCGY$ have areas 16, 16, and 32, respectively. If the squares only intersect pairwise at the vertices A , B , and C , compute the area of the convex hexagon $THINGY$.
- T-9. Compute the coefficient of x^8 in the expansion of $(x^2 + x + 1)^8$ after combining like terms.
- T-10. Consider the following 3×3 grid with its center square removed, as shown.



Compute the number of distinct ways to fill in the grid with the integers 1 through 8, each appearing exactly once, so that:

- In each of the three rows, the entries are increasing from left to right.
- In each of the three columns, the entries are increasing from top to bottom.

T-11. Kevin and Tarsha are playing a game with four fair standard six-sided dice. The four dice are rolled and if the numbers appearing on top of the four dice are all different, Kevin pays Tarsha \$1. If not, Tarsha pays Kevin \$ k dollars. Compute the value of k such that the game is fair, in other words, the expected value of each play of the game to both players is zero.

T-12. Compute the minimum value of ab such that $\log_2(a^4b^{-3}) = 3$ and $\log_2(a^4b^3) = 9$.

T-13. Compute

$$\sum_{n=3}^{10} \frac{20}{(n-2)(n+2)}.$$

T-14. The six-digit number 142857 has the property that moving the rightmost digit to the left of the number results in multiplying the number by five ($142857 \cdot 5 = 714285$). Compute the smallest six-digit number with the property that moving the rightmost digit to the left of the number results in multiplying the number by four.

T-15. Compute the number of distinct sequences $(x_1, x_2, \dots, x_{14})$ with the following properties:

- $x_k \in \{1, 2, \dots, 14\}$ for each $k = 1, 2, \dots, 14$.
- The number $20x_{k+1}^2 - 17x_k^2$ is divisible by 14 for each $k = 1, 2, \dots, 13$.